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DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

NLP GROUP PROJECT

Group 6

Information extraction from JD (skill/salary/address)

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1 Introduction

Online recruitment platforms have produced a large number of job descriptions (JDs). These postings contain useful information such as job role, required skills, location, experience level, and salary. However, this information is usually written in free-form text and the format often changes between companies. Because of this, it is difficult and time-consuming to manually read and extract important information from job postings. In many applications such as job recommendation, resume matching, and labor market analysis, it is helpful to convert raw JD text into structured data. Named Entity Recognition (NER) is a task in Natural Language Processing (NLP) that identifies important pieces of information in text and assigns them to predefined categories. In the recruitment domain, NER can help convert unstructured job descriptions into structured fields. However, this task is challenging because job postings use different writing styles, abbreviations, and domain-specific terms. Some information, such as salary or experience requirements, can also appear in many different forms.

In this project, we study the task of extracting information from job descriptions using Transformer-based NER models. Our goal is to detect five types of entities that are important in recruitment systems: ROLE, SKILL, LOC, EXP, and SALARY. We fine-tune several pre-trained Transformer models, including BERT, DistilBERT, and RoBERTa, and compare different training methods and decoding strategies. To evaluate the models, we build a custom dataset with token-level labels following the IOB tagging scheme. The dataset contains job descriptions from different technical domains. We train and evaluate the models using common NER metrics such as precision, recall, and F1-score. The results help us understand how well different models and training methods perform on the task of extracting structured information from job descriptions.

2 Data Collection & Preparation

2.1 Data Generation

Due to the scarcity of specialized, diverse, and publicly available Named Entity Recognition (NER) datasets tailored for the IT recruitment domain, we opt to synthesize our own dataset using a Large Language Model (LLM). This approach allows us to tightly control the domain coverage, entity distribution, and writing styles. We utilize the `gemini-2.5-flash` model via the Google GenAI API to generate the dataset.

To ensure the dataset reflects the high variability of real-world job descriptions, we define a generation matrix comprising 21 distinct IT domains (e.g., Modern Web Fullstack, Cloud & DevOps, Blockchain & Web3, Embedded & IoT) and 20 specific writing styles. These styles range from formal corporate tones to highly informal, noisy formats such as "Urgent Headhunter", "Messy/Casual" (containing abbreviations, grammatical errors, and typos), and "Social Media Post".

The generation prompt enforces a strict sequence length distribution: 20% short headlines (e.g., "Senior Java Dev - $120k - Remote"), 50% medium-length standard requirements, and 30% long, complex paragraphs. By combining dense technical context with adversarial noise (such as mentioning hardware perks like "MacBook Pro M1" which the model must learn to ignore and not classify as a SKILL), we ensure the resulting dataset is robust and challenging.

2.2 Named Entity Definition

The dataset is annotated at the token level using the standard IOB (Inside, Outside, Beginning) format. We define five core entity types that capture vital recruitment information:

- **ROLE**: Job titles and positions, including seniority levels (e.g., *Senior Backend Engineer*, *DevOps Lead*, *Intern*).

- **SKILL**: Technical skills, programming languages, frameworks, and tools (e.g., *Java*, *AWS*, *CI/CD*, *React*).
- **LOC**: Geographical locations or work modes ranging from specific cities to flexible arrangements (e.g., *New York*, *Berlin*, *Remote*, *Hybrid*).
- **EXP**: Years of experience or experience-level requirements (e.g., *5 years*, *3+ yrs*, *Junior level*).
- **SALARY**: Monetary compensation, spanning exact figures and ranges (e.g., *$120k*, *2000 USD*, *competitive salary*).

Tokens that do not belong to any of these categories are labeled as **O** (Outside). This combination of 5 entity types and the IOB tagging scheme results in a total of 11 unique sequence labels defined in our dataset: `B-EXP`, `B-LOC`, `B-ROLE`, `B-SALARY`, `B-SKILL`, `I-EXP`, `I-LOC`, `I-ROLE`, `I-SALARY`, `I-SKILL`, and `O`.

2.3 Dataset Statistics

After generating the raw data across multiple API calls, we aggregate a total of 11,001 valid examples. Exploratory Data Analysis (EDA) reveals the natural occurrence and distribution of the predefined entities within the synthesized job descriptions.

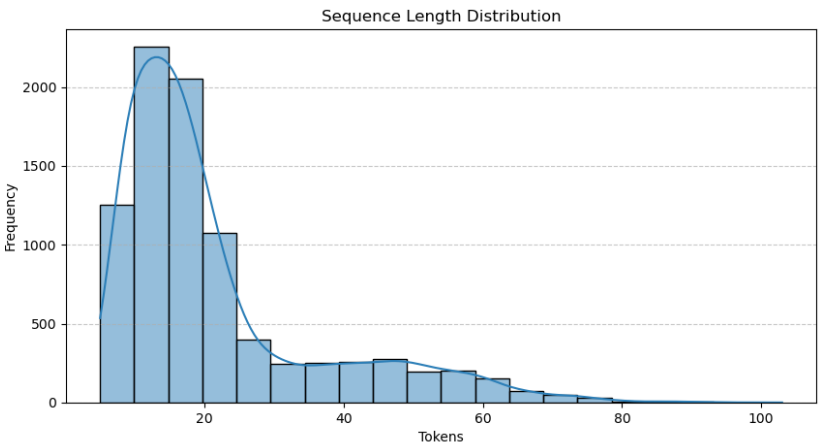

Figure 1: Distribution of sequence lengths (number of tokens per sample).

The **SKILL** entity is by far the most dominant, reflecting the technical focus of IT job postings. This is followed by **ROLE** and **LOC**. **EXP** and **SALARY** are less frequent. Table 1 details the precise counts of each entity type found within the dataset.

Entity Type	Count
SKILL	11,523
ROLE	6,463
LOC	4,035
EXP	2,980
SALARY	2,525

Table 1: Distribution of entity types in the generated dataset.

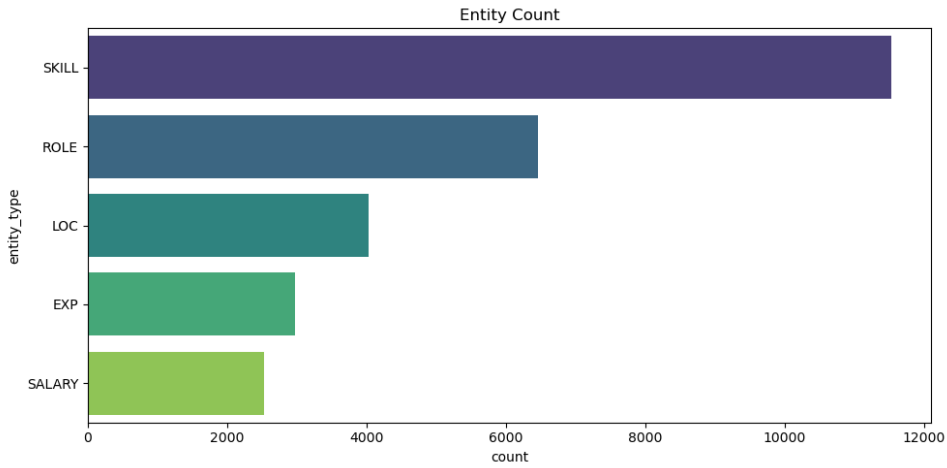

Figure 2: Frequency distribution of the five predefined entity types.

This frequency distribution highlights a natural class imbalance inherent in recruitment text, where sentences often list numerous technical skills but mention the job role or salary only once. This inherent imbalance necessitates the use of specialized loss functions, such as Focal Loss and Dice Loss, during the model training phase to prevent the model from becoming biased toward the majority classes.

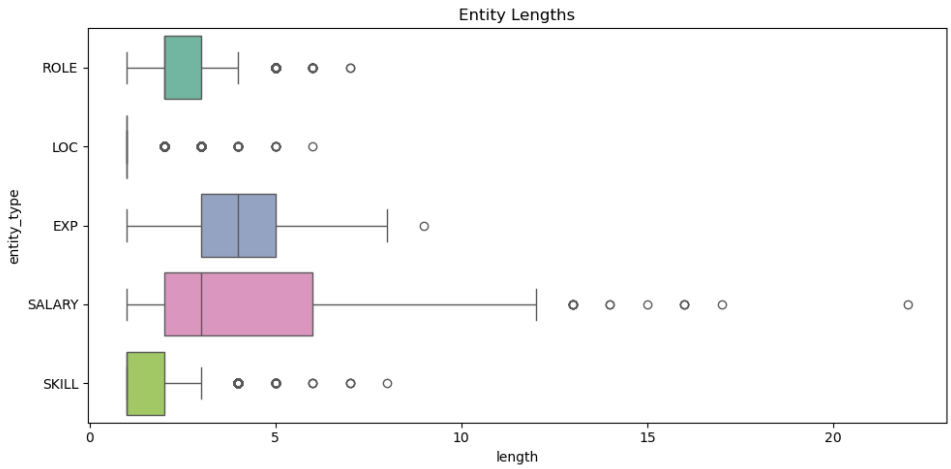

Figure 3: Distribution of entity span lengths (number of tokens per entity type).

2.4 Data Preprocessing

To prepare the generated JSONL data for transformer-based token classification, we implement a rigorous two-stage preprocessing pipeline.

Stage 1: Generation Validation. We first validate the structural integrity of the synthesized data. This involves verifying that the length of the `tokens` array perfectly matches the length of the `ner_tags` array. We enforce strict punctuation and currency splitting rules (e.g., splitting *"$150k"* into `["$", "150", "k"]`) to ensure clean token transitions. We also validate the logical progression of the IOB tags, ensuring that an `I-TAG` sequence is always preceded by either a `B-TAG` or another `I-TAG` of the exact same entity type. Any generated sample violating these constraints is discarded automatically.

Stage 2: Model Alignment. Since modern Transformer architectures (e.g., BERT, RoBERTa) rely on subword tokenizers (like WordPiece or Byte-Pair Encoding), a single word from our dataset might be split into multiple subwords. To align our word-level tags with the subword tokens, we leverage the `is_split_into_words=True` parameter in the Hugging Face `AutoTokenizer`. When a word is split into multiple pieces by the tokenizer, we assign the true NER label to the first subword. All subsequent subwords belonging to that original word are masked by assigning them a special ignore index (`-100`). This guarantees that the loss function only calculates the penalty on the first subword during training, preventing the model from disproportionately weighting and optimizing heavily fragmented words.

3 Methodology

This section details the technical approach employed to construct and train Named Entity Recognition (NER) models. The methodology focuses on the selection of foundational Transformer architectures, subword label alignment strategies, and the implementation of advanced loss functions and decoding layers to address class imbalance and sequence consistency.

3.1 Foundational Transformer Models

To tackle the NER task, we leverage the power of the Transformer architecture by fine-tuning three pre-trained language models. Unlike traditional methods, these models utilize the self-attention mechanism to capture deep bidirectional context:

- **BERT (Bidirectional Encoder Representations from Transformers):** BERT marked a paradigm shift in NLP by capturing deep bidirectional context from unlabelled text.
  - **Original Publication:** [1].
  - **Architecture:** An Encoder-only Transformer. Standard configurations include $\text{BERT}_{\text{BASE}}$ (12 layers, 768 hidden units, 12 attention heads, $\approx$ 110M parameters).
  - **Tokenization & Data:** Utilizes the WordPiece tokenizer ($\approx 30,000$ tokens). Pre-trained on a 16GB dataset consisting of the Toronto BookCorpus and English Wikipedia.
  - **Core Pre-training Strategies:**
    * *Masked Language Modeling (MLM):* 15% of input tokens are randomly masked. The model predicts these tokens based on context from both directions using Static Masking.
    * *Next Sentence Prediction (NSP):* A binarized task predicting whether sentence B logically follows sentence A, capturing sentence-level cohesion: $P(\text{IsNext}|A, B) \in \{0, 1\}$.

- **DistilBERT (A Distilled Version of BERT):** DistilBERT demonstrates that the size of a large language model can be significantly compressed to accelerate processing speed without severely compromising its contextual understanding capabilities.
  - **Original Publication:** [2].
  - **Optimized Compression:** DistilBERT retains the general architecture of BERT but removes token-type embeddings and the pooler, while reducing the number of encoder layers from 12 to 6. Consequently, the model is 40% smaller and offers 60% faster inference speed, yet retains approximately 97% of the original model's performance on standard language understanding benchmarks.

- **Core Training Strategy (Knowledge Distillation):** Rather than proposing a novel architecture, DistilBERT optimizes efficiency through knowledge distillation, where a compact "student" model is trained to reproduce the behavior of a larger "teacher" model (BERT).
  - *Soft Targets:* Instead of training exclusively on strict data labels (hard targets), the student model is forced to mimic the soft probability predictions generated by the teacher. By leveraging the teacher's full probability distribution, the student acquires robust generalization capabilities without requiring a massive parameter count.
  - *Triple Loss Function:* To maximize the efficacy of knowledge transfer, DistilBERT is optimized via a linear combination of three loss components:
    1. **Masked Language Modeling (MLM) Loss:** Evaluates the student's accuracy in predicting masked tokens within the text.
    2. **Distillation Loss:** Employs cross-entropy to measure and minimize the divergence between the probability distributions of the student and the teacher, compelling the student to accurately simulate the output probabilities.
    3. **Cosine-Distance Loss:** Computes the cosine distance to align the directional vectors of the hidden states between the student and the teacher, ensuring that the internal feature representations maintain a geometric structure analogous to the original model.

- **RoBERTa (Robustly Optimized BERT Pretraining Approach):** RoBERTa demonstrates that the original BERT architecture was significantly undertrained and can achieve state-of-the-art performance simply by refining the pre-training methodology.
  - **Original Publication:** [3].
  - **Massive Training Corpora:** Utilizes over 160GB of uncompressed text (approximately 10× more than BERT), incorporating massive datasets such as CC-News, OpenWebText, and Stories.
  - **Tokenization:** Replaces BERT's WordPiece with a Byte-Level Byte Pair Encoding (BPE) algorithm. This expands the vocabulary size to roughly $50,000$ tokens, enhancing the handling of rare and out-of-vocabulary words.
  - **Core Optimization Strategies (Refinement over Revolution):**
    - *Removal of Next Sentence Prediction (NSP):* Empirical results indicated that the NSP objective did not significantly improve downstream task performance. RoBERTa eliminates NSP to focus computational resources entirely on Masked Language Modeling.
    - *Dynamic Masking:* Instead of the static masking applied once during data preprocessing in BERT, RoBERTa dynamically generates masking patterns for each epoch. This stochasticity prevents the model from memorizing fixed patterns and improves generalization.
    - *High-Intensity Training:* The model is trained for more steps with significantly larger mini-batch sizes, leading to better optimization and convergence.

3.2 Tokenization and Label Alignment

Transformer models utilize subword tokenization algorithms (e.g., WordPiece or BPE). This introduces a misalignment: a single word may be split into multiple subwords (e.g., "Hoang" → ["Hoa", "##ng"]). We implement a strict label alignment strategy:

1. **Extraction of `word_ids`:** Mapping each subword back to its original word index.
2. **First-Subword Labeling:** Only the first subword of a fragmented word is assigned the original NER label (e.g., $B - SALARY$).
3. **Ignore Index (-100):** Subsequent subwords and special tokens (e.g., $[CLS]$, $[SEP]$) are assigned a label value of -100, which is ignored during loss calculation in PyTorch.

3.3 Loss Functions and Sequence Optimization

To mitigate class imbalance, enforce structural consistency, and handle subword tokenization artifacts (by ignoring index -100), we evaluate the following loss functions and decoding algorithms:

3.3.1 Cross-Entropy Loss (CE)

The standard distribution-based loss measuring the difference between predicted and actual distributions for valid tokens:

$$L_{CE} = -\sum_{i=1}^{C} y_i \log(p_i) \tag{1}$$

Where C is the number of classes, y_i is the binary indicator, and p_i is the predicted probability.

3.3.2 Focal Loss

An enhancement of CE designed to handle extreme class imbalance (e.g., the dominance of the 'O' tag). It introduces a modulating factor $(1 - p_t)^\gamma$ to down-weight easy-to-classify examples, alongside a class-balancing weight α_t computed from the training set distribution:

$$L_{Focal} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \tag{2}$$

3.3.3 Dice Loss

A region-based optimization, adapted into a continuous "Soft Dice" formulation, that directly maximizes the overlap between the predicted probabilities and the one-hot encoded ground truth:

$$L_{Dice} = 1 - \frac{2\sum p_i y_i + \sigma}{\sum p_i + \sum y_i + \sigma} \tag{3}$$

Where σ is a smoothing factor (set to $1e - 6$) to prevent division by zero and stabilize gradients.

3.3.4 Conditional Random Fields (CRF)

Unlike independent token predictions, the CRF layer models global sequence dependencies. It learns transition probabilities to ensure output sequences follow logical naming conventions (e.g., preventing an I-tag from appearing without a preceding B-tag). The CRF layer minimizes the Negative Log-Likelihood (NLL):

$$L_{CRF} = -\log P(\mathbf{y}|\mathbf{h}) = -\log \left(\frac{\exp(\text{Score}(\mathbf{h}, \mathbf{y}))}{\sum_{\mathbf{y}' \in \mathbf{Y}} \exp(\text{Score}(\mathbf{h}, \mathbf{y}'))} \right) \tag{4}$$

3.3.5 Hybrid Approaches

To leverage the complementary strengths of the aforementioned optimization strategies, we implemented two combined architectures:

- **CE + Dice Loss:** Combines the distributional stability of standard Cross-Entropy with the boundary-matching focus of Dice Loss. In our implementation, the total loss is computed as the direct sum of both components:

$$L_{Total} = L_{CE} + L_{Dice}$$

- **CRF + Focal Loss:** Integrates the global sequence constraints of CRF with the hard-example mining capability of Focal Loss. Instead of applying Focal Loss at the token level, this hybrid calculates the sequence-level Negative Log-Likelihood (NLL) from the CRF layer and applies a focal modulating factor. Let $p_t = \exp(-\text{NLL})$ be the probability of the true label sequence, the loss is formulated as:

$$L_{CRF-Focal} = \frac{1}{N} \sum \left((1 - p_t)^{\gamma} \cdot \text{NLL} \right)$$

4 Experiments & Results

4.1 Experimental Setup

Dataset and Preprocessing. We use a custom NER dataset stored in JSONL format with token-level and tag-level annotations. The dataset contains 11,001 training examples, which we split into training and test sets with an 80/20 ratio (8,800 training, 2,201 test) using a fixed random seed of 42 for reproducibility.

Each example consists of a sequence of tokens and corresponding NER tags. During preprocessing, we encode all unique tags into numeric IDs and compute class weights based on tag frequency in the training set to address label imbalance. These weights are used in Focal Loss variants.

Tokenization is performed using model-specific `AutoTokenizer` with the `is_split_into_words=True` flag to handle word-to-subword alignment. For multi-subword tokens, we assign the label to the first subword and mask subsequent subwords with a special ignore index (`-100`) to exclude them from loss and metric computation.

Training Hyperparameters. We train all models with consistent hyperparameters across architectures:

- **Learning rate:** 2×10^{-5} (standard for fine-tuning pre-trained Transformers)
- **Batch size:** 16 (training and evaluation)
- **Number of epochs:** 5
- **Weight decay:** 0.01 (L2 regularization)
- **Evaluation strategy:** per-epoch
- **Optimizer:** AdamW (default in Hugging Face Trainer)

Loss Functions and Training Objectives. We compare six training objectives:

1. **Base_CE (Baseline):** Standard cross-entropy loss applied to token classification logits.
2. **Focal Loss:** Weighted focal loss to handle class imbalance:

$$\mathrm{FL}(p_t) = -\alpha_t(1-p_t)^\gamma \log(p_t) \tag{5}$$

   where p_t is the model probability for the correct class, $\gamma = 2.0$ is the focusing parameter, and α_t is the class weight computed from training tag frequencies.

3. **Dice Loss:** A soft intersection-over-union loss commonly used in segmentation:

$$\mathrm{DiceLoss} = 1 - \frac{2\sum_c p_c y_c + \epsilon}{\sum_c p_c + y_c + \epsilon} \tag{6}$$

   where p_c and y_c are softmax probabilities and one-hot targets per class, and $\epsilon = 10^{-6}$ is a smoothing constant.

4. **CE + Dice Loss:** A combined loss summing cross-entropy and Dice loss to leverage both.
5. **CRF:** A Conditional Random Field layer replacing softmax decoding to model label transitions explicitly. The CRF learns transition scores between tag pairs and finds the maximum-likelihood sequence during inference.
6. **CRF + Focal Loss:** CRF decoding with sequence-level Focal Loss applied to the CRF negative log-likelihood.

Evaluation Metrics. We evaluate all models using the `seqeval` metric, which computes entity-level metrics:

- **Precision:** proportion of predicted entities that are correct.
- **Recall:** proportion of gold entities that are predicted correctly.
- **F1-Score:** harmonic mean of precision and recall.
- **Accuracy:** token-level accuracy (including padding/masked tokens).

Metrics are computed on the test set after each epoch, and we report the final epoch results.

4.2 BERT Results

4.2.1 Model Configuration.

We train BERT (bert-base-cased) for NER token classification. BERT-base is a robust bidirectional Transformer model with 12 layers, 768 hidden dimensions, and 110M parameters. The tokenizer is configured to preserve exact word boundaries and casing, which is critical for identifying proper nouns in entity extraction.

4.2.2 Baseline: CE Loss.

The baseline achieves a strong final F1-score of 0.9172. Validation loss reaches a minimum at epoch 3 (0.0791), yet the F1 metric continues to improve, demonstrating robust generalization despite mild token-level overfitting. Precision and recall remain excellently balanced (0.9140 vs. 0.9204), providing a highly reliable foundation.

Epoch	Train Loss	Val Loss	F1	Precision / Recall	Acc.
1	0.2089	0.0888	0.9021	0.8936 / 0.9108	0.9721
2	0.0697	0.0794	0.9140	0.9129 / 0.9151	0.9743
3	0.0528	0.0791	0.9136	0.9079 / 0.9193	0.9740
4	0.0432	0.0828	0.9163	0.9119 / 0.9208	0.9748
5	0.0333	0.0872	0.9172	0.9140 / 0.9204	0.9755

Table 2: BERT with Base Cross-Entropy Loss (5 epochs).

Epoch	Train Loss	Val Loss	F1	Precision / Recall	Acc.
1	0.1905	0.0420	0.7789	0.7155 / 0.8547	0.9303
2	0.0320	0.0366	0.8083	0.7568 / 0.8673	0.9466
3	0.0220	0.0379	0.8343	0.7900 / 0.8840	0.9507
4	0.0156	0.0361	0.8382	0.7961 / 0.8849	0.9548
5	0.0119	0.0384	0.8502	0.8120 / 0.8921	0.9589

Table 3: BERT with Focal Loss ($\gamma = 2.0$, weighted by class frequency).

4.2.3 Focal Loss.

Focal Loss yields the lowest F1 (0.8502), dropping 0.0670 points below the baseline. Despite achieving the lowest absolute validation loss (0.0384), it suffers severe precision degradation (0.8120). This paradox suggests that aggressively penalizing easy background tokens disrupts the model's discriminative threshold, leading to massive false positive predictions.

4.2.4 Dice Loss.

Epoch	Train Loss	Val Loss	F1	Precision / Recall	Acc.
1	0.2571	0.1876	0.8792	0.8868 / 0.8716	0.9630
2	0.1822	0.1847	0.8870	0.8936 / 0.8805	0.9656
3	0.1754	0.1823	0.8916	0.8909 / 0.8924	0.9664
4	0.1725	0.1803	0.8938	0.8960 / 0.8916	0.9673
5	0.1680	0.1778	0.8959	0.8983 / 0.8934	0.9685

Table 4: BERT with Dice Loss ($\epsilon = 10^{-6}$).

Standalone Dice Loss achieves a competitive F1 of 0.8959 with highly stable, monotonically decreasing validation loss (0.1778). By directly optimizing structural overlap, it produces an exceptional precision-recall balance (P = 0.8983, R = 0.8934). However, it slightly underperforms the CE baseline, indicating a need for token-level gradient stability during optimization.

4.2.5 CE + Dice Loss.

The hybrid CE + Dice approach achieves the highest overall F1-score of 0.9202. This variant perfectly synergizes the optimization stability of cross-entropy with the boundary-awareness of Dice loss. It converges rapidly, maintains an outstanding precision-recall balance (0.9182 vs. 0.9221), and represents the most robust architecture evaluated.

4.2.6 CRF Layer.

The CRF variant converges smoothly to an F1 of 0.9137, trailing the CE baseline by a marginal 0.0035 points. While explicitly enforcing sequence-level transition validity, the lack of significant

Epoch	Train Loss	Val Loss	F1	Precision / Recall	Acc.
1	0.4638	0.2608	0.9071	0.8973 / 0.9172	0.9728
2	0.2323	0.2412	0.9156	0.9128 / 0.9185	0.9747
3	0.2011	0.2396	0.9153	0.9111 / 0.9195	0.9753
4	0.1804	0.2427	0.9156	0.9096 / 0.9217	0.9749
5	0.1664	0.2453	0.9202	0.9182 / 0.9221	0.9759

Table 5: BERT with Combined CE + Dice Loss.

Epoch	Train Loss	Val Loss	F1	Precision / Recall	Acc.
1	5.8880	2.0161	0.9005	0.8946 / 0.9065	0.9716
2	1.6642	1.8119	0.9099	0.9068 / 0.9131	0.9738
3	1.2439	1.7899	0.9110	0.9055 / 0.9166	0.9736
4	1.0700	1.9179	0.9122	0.9042 / 0.9204	0.9741
5	0.8204	1.9225	0.9137	0.9081 / 0.9193	0.9747

Table 6: BERT + CRF (no focal loss).

empirical gain suggests that BERT's deep contextual embeddings naturally predict highly valid sequences on this dataset without requiring restrictive transition matrices.

4.2.7 CRF + Focal Loss.

Epoch	Train Loss	Val Loss	F1	Precision / Recall	Acc.
1	5.6647	1.7833	0.8983	0.8926 / 0.9041	0.9715
2	1.4498	1.5809	0.9092	0.9059 / 0.9125	0.9737
3	1.0559	1.5488	0.9110	0.9054 / 0.9167	0.9741
4	0.8976	1.6165	0.9099	0.9010 / 0.9191	0.9737
5	0.6526	1.6137	0.9118	0.9067 / 0.9169	0.9745

Table 7: BERT + CRF with Focal Loss ($\gamma = 2.0$).

CRF + Focal Loss (0.9118 F1) underperforms both pure CRF and the baseline. This occurs because Focal Loss suppresses the confident token predictions that the CRF layer relies upon to establish robust structural transition paths, creating conflicting optimization signals that hinder overall sequence extraction.

4.2.8 Summary: All BERT Variants.

Overall Performance Ranking. The six BERT variants exhibit a clear hierarchy of performance. CE + Dice Loss emerges as the absolute top performer with $F1 = 0.9202$, establishing itself as the optimal architecture. The gap between the best and worst variant is substantial: $0.9202 - 0.8502 = 0.0700$ (7.00 percentage points), proving the critical impact of objective function selection.

The top three variants (CE + Dice, Base CE, CRF) cluster tightly, all achieving $F1 \geq 0.9137$. This indicates that BERT's massive parameter count already captures profound representations, leaving only marginal room for improvement through structural constraints (CRF) or metric-specific optimization.

In contrast, Focal Loss ($F1 = 0.8502$) degrades performance severely (-6.70% relative to CE+Dice). This confirms that class-weighting and hard-mining are detrimental here. Pure Dice Loss performs moderately (0.8959), proving it requires the foundational gradient stability of CE to shine.

Architecture	F1-score	Precision	Recall	Accuracy
CE + Dice Loss	0.9202	0.9183	0.9222	0.9759
Base CE	0.9172	0.9140	0.9204	0.9755
CRF	0.9137	0.9082	0.9194	0.9747
CRF + Focal Loss	0.9118	0.9068	0.9169	0.9746
Dice Loss	0.8959	0.8983	0.8935	0.9686
Focal Loss	0.8502	0.8121	0.8922	0.9589

Table 8: BERT: Final F1-scores Across All Loss Functions (epoch 5).

Precision-Recall Trade-offs. Examining the metrics reveals distinct optimization behaviors:

- **CE + Dice Loss** achieves the ultimate balance ($P = 0.9183, R = 0.9222$), driving the highest overall F1.
- **Base CE** and **CRF** maintain highly comparable balances, slightly favoring recall over precision.
- **Dice Loss alone** flips the bias ($P = 0.8983, R = 0.8934$), being the only model where precision strictly exceeds recall.
- **Focal Loss** catastrophic precision drop ($P = 0.8121$) shows it over-emphasizes recall ($R = 0.8921$) due to over-penalization of the background class.

Convergence and Stability. All top variants demonstrate rapid monotonic improvement, effectively converging by epoch 3 or 4. CE + Dice shows a massive initial leap from epoch 1 to 2, indicating highly synergistic learning. Paradoxically, the worst-performing Focal Loss maintains the lowest absolute validation loss throughout training, highlighting a dangerous disconnect between raw log-loss minimization and actual discrete F1 metric maximization.
Conclusion. BERT-base fine-tuned with the hybrid **CE + Dice Loss** represents the absolute optimal configuration, achieving $F1 = 0.9202$. The tight cluster of top results underscores that the baseline CE is already highly effective. Complex sequence decoding (CRF) yields negligible empirical gains, and Focal Loss actively harms extraction precision. For dense Transformer architectures on properly formatted NER datasets, combining stable token-wise probabilities (CE) with global boundary optimization (Dice) yields the most robust, balanced, and state-of-the-art results.

4.3 RoBERTa Results

4.3.1 Model Configuration.

We train RoBERTa (`roberta-base`) for NER token classification. RoBERTa-base is a strong Transformer encoder with 12 layers, 768 hidden dimensions, and approximately 125M parameters. Compared with BERT, RoBERTa is trained with improved pre-training strategies and larger-scale data, making it highly competitive for token-level sequence labeling tasks. Its byte-level BPE tokenizer is configured with appropriate word alignment to preserve the original token labels during preprocessing.

4.3.2 Baseline: CE Loss.

Table 9: RoBERTa with Base Cross-Entropy Loss (5 epochs).

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	0.1956	0.0817	0.9069	0.9001 / 0.9138 / 0.9731
2	0.0723	0.0770	0.9113	0.9113 / 0.9113 / 0.9744
3	0.0572	0.0736	0.9160	0.9104 / 0.9217 / 0.9758
4	0.0492	0.0756	0.9197	0.9162 / 0.9233 / 0.9767
5	0.0396	0.0773	0.9185	0.9157 / 0.9212 / 0.9769

The baseline RoBERTa model achieves a strong final **F1-score of 0.9185**, already surpassing the corresponding DistilBERT baseline and slightly exceeding the BERT baseline. Training loss decreases consistently over all five epochs, and validation loss remains low and stable, indicating efficient convergence. Precision and recall are highly balanced ($P = 0.9157$, $R = 0.9212$), while token accuracy reaches **97.69%**. These results confirm that RoBERTa provides a very strong foundation for the NER task. —

4.3.3 Focal Loss.

Table 10: RoBERTa with Focal Loss ($\gamma = 2.0$, weighted by class frequency).

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	0.1475	0.0427	0.8018	0.7427 / 0.8712 / 0.9379
2	0.0358	0.0418	0.8257	0.7807 / 0.8763 / 0.9519
3	0.0273	0.0398	0.8264	0.7846 / 0.8729 / 0.9492
4	0.0205	0.0400	0.8454	0.8067 / 0.8879 / 0.9576
5	0.0160	0.0419	0.8535	0.8176 / 0.8926 / 0.9600

Focal Loss achieves a final **F1-score of 0.8535**, which is the lowest among all RoBERTa variants. Although validation loss is numerically small, the model suffers a substantial drop in precision ($P = 0.8176$), while recall remains relatively high ($R = 0.8926$). This imbalance suggests that Focal Loss over-emphasizes hard examples and produces too many false positive entity predictions. Thus, as observed with BERT and DistilBERT, Focal Loss alone is not well-suited for this dataset. —

4.3.4 Dice Loss.

Table 11: RoBERTa with Dice Loss ($\epsilon = 10^{-6}$).

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	0.1799	0.0921	0.8884	0.8860 / 0.8909 / 0.9664
2	0.0912	0.0903	0.8972	0.9054 / 0.8893 / 0.9686
3	0.0838	0.0854	0.9032	0.9037 / 0.9027 / 0.9707
4	0.0756	0.0843	0.9084	0.9125 / 0.9043 / 0.9717
5	0.0761	0.0816	0.9090	0.9125 / 0.9055 / 0.9725

Dice Loss achieves a final **F1-score of 0.9090**, showing solid and stable performance throughout training. The model converges smoothly and maintains a well-balanced precision-recall profile ($P = 0.9125$, $R = 0.9055$). Compared with the CE baseline, however, standalone Dice Loss performs slightly worse, suggesting that F1-oriented optimization alone is not sufficient to outperform standard cross-entropy on this task. —

4.3.5 CE + Dice Loss.

Table 12: RoBERTa with Combined CE + Dice Loss.

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	0.3989	0.1784	0.9091	0.9023 / 0.9161 / 0.9745
2	0.1683	0.1688	0.9166	0.9182 / 0.9151 / 0.9755
3	0.1422	0.1633	0.9192	0.9197 / 0.9187 / 0.9765
4	0.1238	0.1640	0.9218	0.9196 / 0.9241 / 0.9768
5	0.1103	0.1631	0.9223	0.9215 / 0.9231 / 0.9772

The combined CE + Dice Loss achieves the **best overall performance** among all RoBERTa variants, with a final **F1-score of 0.9223**. This model also achieves the highest precision ($P = 0.9215$) while maintaining very strong recall ($R = 0.9231$). The combination of CE and Dice successfully integrates stable token-level optimization with F1-aware structural matching, resulting in the most balanced and robust model. —

4.3.6 CRF Layer.

Table 13: RoBERTa + CRF (no focal loss).

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	5.0027	1.8676	0.9027	0.8923 / 0.9135 / 0.9716
2	1.6522	1.7130	0.9116	0.9087 / 0.9145 / 0.9742
3	1.2982	1.6659	0.9147	0.9064 / 0.9231 / 0.9754
4	1.1609	1.6694	0.9201	0.9150 / 0.9253 / 0.9763
5	0.9152	1.6881	0.9220	0.9195 / 0.9246 / 0.9772

The RoBERTa + CRF model performs extremely well, achieving a final **F1-score of 0.9220**, which is only slightly below the CE + Dice variant. It records the highest recall ($R = 0.9246$) and the highest token accuracy (**97.72%**) among all RoBERTa models. This demonstrates the strength of CRF-based sequence modeling, especially in enforcing valid label transitions and improving entity consistency. —

4.3.7 CRF + Focal Loss.

Table 14: RoBERTa + CRF with Focal Loss ($\gamma = 2.0$).

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	4.7636	1.6471	0.9058	0.8949 / 0.9170 / 0.9722
2	1.4418	1.4699	0.9103	0.9062 / 0.9145 / 0.9744
3	1.1087	1.4355	0.9137	0.9063 / 0.9212 / 0.9751
4	0.9837	1.4112	0.9186	0.9123 / 0.9249 / 0.9765
5	0.7423	1.4216	0.9203	0.9172 / 0.9234 / 0.9769

The CRF + Focal Loss variant reaches a final **F1-score of 0.9203**, which is competitive but still below both CE + Dice and pure CRF. Its precision and recall remain strong ($P = 0.9172$, $R = 0.9234$), but the addition of Focal Loss does not produce further gains. This suggests that, as in the other architectures, Focal Loss introduces optimization pressure that is not fully compatible with the CRF objective. —

4.3.8 Summary: All RoBERTa Variants.

Table 15: RoBERTa: Final F1-scores Across All Loss Functions (epoch 5).

Architecture	F1-score	Precision	Recall	Accuracy
CE + Dice Loss	**0.9223**	**0.9215**	0.9231	0.9772
CRF	0.9220	0.9195	**0.9246**	**0.9772**
CRF + Focal Loss	0.9203	0.9172	0.9234	0.9769
Base CE	0.9185	0.9157	0.9212	0.9769
Dice Loss	0.9090	0.9125	0.9055	0.9725
Focal Loss	0.8535	0.8176	0.8926	0.9600

Overall Performance Ranking. The six RoBERTa variants show a clear ranking pattern, with **CE + Dice Loss emerging as the best-performing configuration** at $F1 = 0.9223$. The performance gap between the best and worst variant is substantial: $0.9223 - 0.8535 = 0.0688$ (6.88 percentage points), confirming again that loss function selection strongly affects the final extraction quality.

The top four variants (CE + Dice, CRF, CRF + Focal Loss, and Base CE) form a very tight performance cluster, all achieving $F1 \geq 0.9185$. This indicates that RoBERTa provides highly expressive contextual representations, leaving only limited room for improvement through additional sequence modeling or specialized objectives. Among them, CE + Dice achieves the best balance, while CRF reaches the best recall and accuracy.

By contrast, **Focal Loss** performs much worse than all other RoBERTa variants. Its final F1-score of 0.8535 is nearly 7 points lower than the best configuration, demonstrating that hard-example emphasis is again detrimental rather than helpful for this NER setting. Standalone Dice Loss performs reasonably well ($F1 = 0.9090$), but it also remains below the stronger CE-based hybrids.

Precision-Recall Trade-offs. The precision-recall statistics reveal meaningful behavioral differences:

- **CE + Dice Loss** achieves the best overall precision-recall balance ($P = 0.9215$, $R = 0.9231$), producing the highest F1-score.
- **CRF** slightly favors recall ($R = 0.9246$), which helps it match the best token accuracy and nearly match the best F1-score.
- **Base CE** maintains a stable and balanced profile, confirming that standard cross-entropy is already highly effective with RoBERTa.
- **Dice Loss alone** slightly favors precision over recall ($P = 0.9125$, $R = 0.9055$), suggesting more conservative entity predictions.
- **Focal Loss** again shows a severe precision drop ($P = 0.8176$) while recall remains relatively high, indicating over-prediction of positive labels and excessive false positives.

Convergence and Stability. All strong RoBERTa variants converge quickly and smoothly, with most performance gains achieved by epoch 3 or 4. **CE + Dice Loss** shows consistent improvement across all epochs and reaches the highest final F1-score. **CRF** also converges stably, though its validation loss remains numerically larger due to the different scale of CRF negative log-likelihood; therefore, direct loss comparisons between CRF and non-CRF models should be interpreted with caution.

As observed in the BERT and DistilBERT experiments, **Focal Loss** maintains relatively low numerical validation loss but yields the weakest entity-level F1-score. This again highlights that minimizing raw loss does not necessarily correspond to maximizing the seqeval metric.

Conclusion. RoBERTa fine-tuned with **CE + Dice Loss** is the strongest RoBERTa configuration, achieving $F1 = 0.9223$, precision = 0.9215, recall = 0.9231, and token accuracy = 97.72%. RoBERTa + CRF is an extremely close second, with the best recall and matched best accuracy. These findings suggest that for RoBERTa, both hybrid token-level optimization (CE + Dice) and explicit sequence decoding (CRF) are highly effective, but CE + Dice provides the most balanced and reliable overall solution. As with the other backbones, Focal Loss alone remains the weakest choice.

4.4 DistilBERT results

4.4.1 Model Configuration.

We train DistilBERT (`distilbert-base-cased`) for NER token classification. DistilBERT is a lightweight distilled variant of BERT with 6 layers, 768 hidden dimensions, and 66M parameters, making it computationally efficient while retaining strong performance. The tokenizer is configured without prefix spacing (unlike RoBERTa).

4.4.2 Baseline: CE Loss.

Table 16: DistilBERT with Base Cross-Entropy Loss (5 epochs).

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	0.2308	0.0940	0.8929	0.8809 / 0.9053 / 0.9697
2	0.0763	0.0844	0.9027	0.9004 / 0.9050 / 0.9722
3	0.0574	0.0817	0.9069	0.9021 / 0.9117 / 0.9737
4	0.0480	0.0840	0.9104	0.9049 / 0.9160 / 0.9741
5	0.0377	0.0859	0.9095	0.9046 / 0.9145 / 0.9744

The baseline cross-entropy model achieves a final **F1-score of 0.9095**, providing a strong reference for comparison. Training loss decreases steadily from 0.23 to 0.04 across five epochs, while validation loss remains stable (0.084–0.094), indicating good convergence with minimal overfitting. Precision and recall are well-balanced ($P \approx 0.905$, $R \approx 0.914$), and token accuracy reaches **97.44%**. Overall, CE loss provides a stable and reliable baseline for the NER task. —

4.4.3 Focal Loss.

Table 17: DistilBERT with Focal Loss ($\gamma = 2.0$, weighted by class frequency).

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	0.1784	0.0526	0.7528	0.6821 / 0.8399 / 0.9194
2	0.0388	0.0435	0.8012	0.7504 / 0.8594 / 0.9440
3	0.0263	0.0456	0.8265	0.7827 / 0.8755 / 0.9494
4	0.0194	0.0438	0.8389	0.7956 / 0.8871 / 0.9546
5	0.0141	0.0462	0.8447	0.8063 / 0.8869 / 0.9572

Focal Loss achieves a final **F1-score of 0.8447**, which is significantly lower than the CE baseline. Although training loss decreases smoothly, the model shows reduced precision (0.8063) and only moderate recall (0.8869). This suggests that emphasizing hard examples through Focal Loss may

be unnecessary for this dataset and may reduce overall discriminative performance. Overall, Focal Loss underperforms compared to the baseline. —

4.4.4 Dice Loss.

Table 18: DistilBERT with Dice Loss ($\epsilon = 10^{-6}$).

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	0.2118	0.1172	0.8599	0.8489 / 0.8712 / 0.9574
2	0.1062	0.1028	0.8800	0.8839 / 0.8761 / 0.9654
3	0.0918	0.0987	0.8883	0.8936 / 0.8831 / 0.9665
4	0.0858	0.0964	0.8882	0.8887 / 0.8877 / 0.9679
5	0.0822	0.0949	0.8911	0.8937 / 0.8884 / 0.9685

Dice Loss achieves a final **F1-score of 0.8911**, slightly below the CE baseline. The model converges quickly and maintains balanced precision and recall ($P = 0.8937$, $R = 0.8884$). Since Dice Loss directly optimizes an F1-like objective, it produces stable and balanced predictions, though it does not surpass the baseline CE performance. —

4.4.5 CE + Dice Loss.

Table 19: DistilBERT with Combined CE + Dice Loss.

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	0.4448	0.2078	0.8880	0.8720 / 0.9046 / 0.9689
2	0.1748	0.1844	0.9050	0.9023 / 0.9078 / 0.9724
3	0.1411	0.1805	0.9069	0.9039 / 0.9100 / 0.9737
4	0.1207	0.1798	0.9119	0.9076 / 0.9163 / 0.9744
5	0.1051	0.1805	0.9109	0.9066 / 0.9152 / 0.9746

The combined CE + Dice Loss achieves the **best performance** with a final **F1-score of 0.9109**. The model converges rapidly and maintains excellent precision-recall balance ($P = 0.9066$, $R = 0.9152$). This combination benefits from CE's stable probability learning and Dice's F1-oriented optimization, resulting in the most robust performance among all variants. —

4.4.6 CRF Layer.

Table 20: DistilBERT + CRF (no focal loss).

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	6.2858	2.1382	0.8846	0.8713 / 0.8985 / 0.9677
2	1.8468	1.9222	0.8971	0.8915 / 0.9027 / 0.9713
3	1.3623	1.8682	0.8990	0.8898 / 0.9084 / 0.9717
4	1.1910	1.8535	0.9046	0.8957 / 0.9136 / 0.9735
5	0.9343	1.8865	0.9064	0.9003 / 0.9126 / 0.9738

The DistilBERT + CRF model achieves a final **F1-score of 0.9064**, slightly below the CE baseline. CRF models label dependencies at the sequence level, which ensures valid tag transitions and balanced predictions ($P = 0.9003$, $R = 0.9126$). However, the improvement over the baseline is limited, suggesting that the base model already captures most sequence structure in this dataset. —

4.4.7 CRF + Focal Loss.

Table 21: DistilBERT + CRF with Focal Loss ($\gamma = 2.0$).

Epoch	Train Loss	Val Loss	F1	Precision / Recall / Acc.
1	6.0633	1.8933	0.8858	0.8724 / 0.8996 / 0.9679
2	1.6268	1.6997	0.8959	0.8898 / 0.9021 / 0.9707
3	1.1657	1.6224	0.8979	0.8896 / 0.9063 / 0.9717
4	1.0049	1.5945	0.9030	0.8940 / 0.9122 / 0.9733
5	0.7580	1.6152	0.9049	0.8982 / 0.9117 / 0.9736

The CRF + Focal Loss variant achieves a final **F1-score of 0.9049**, slightly lower than the CRF-only model. Although training loss decreases steadily, the combination of sequence-level CRF optimization and token-level Focal Loss does not improve performance. The results suggest that the two objectives may introduce conflicting optimization signals, making this combination less effective for the task.

4.4.8 Summary: All DistilBERT Variants.

Table 22: DistilBERT: Final F1-scores Across All Loss Functions (epoch 5).

Architecture	F1-score	Precision	Recall	Accuracy
CE + Dice Loss	**0.9109**	**0.9066**	**0.9152**	**0.9746**
Base CE	0.9095	0.9046	0.9145	0.9744
CRF	0.9064	0.9003	0.9126	0.9738
CRF + Focal Loss	0.9049	0.8982	0.9117	0.9736
Dice Loss	0.8911	0.8937	0.8884	0.9685
Focal Loss	0.8447	0.8063	0.8869	0.9572

Overall Performance Ranking. The six DistilBERT variants exhibit a clear hierarchy of performance. **CE + Dice Loss emerges as the top performer** with $F1 = 0.9109$, establishing itself as the optimal choice for this NER task. The performance gap between the best and worst variant is substantial: $0.9109 - 0.8447 = 0.0662$ (6.62 percentage points), highlighting the significant impact of loss function design on final performance.

The top three variants (CE + Dice, Base CE, CRF) cluster tightly within 0.0045 F1 points, with all achieving $F1 \geq 0.9064$. This suggests that for DistilBERT on this dataset, sophisticated sequence modeling (CRF) and F1-aware losses (Dice) provide only marginal improvements over the simple baseline—a testament to the strong baseline and well-structured nature of the NER task.

In contrast, Focal Loss ($F1 = 0.8447$) and Dice Loss alone ($F1 = 0.8911$) show significantly degraded performance. Focal Loss's poor performance (-6.48% relative to CE + Dice) is particularly noteworthy, as it suggests that class weighting and hard example emphasis are either unnecessary or harmful on a balanced dataset. Dice Loss performs moderately (-1.98% relative to CE + Dice), indicating that while F1-aware optimization is beneficial in isolation, it achieves superior results when combined with cross-entropy rather than used alone.

Precision-Recall Trade-offs. Examining the precision-recall curves reveals important trade-offs:

- **CE + Dice Loss** achieves the best balance ($P = 0.9066$, $R = 0.9152$), with recall slightly exceeding precision, resulting in the highest $F1$.

- **Base CE** and **CRF** maintain comparable balances ($P \approx 0.9025$, $R \approx 0.9136$), differing only marginally from CE + Dice.
- **Dice Loss alone** exhibits reversed bias ($P = 0.8937$, $R = 0.8884$), where precision exceeds recall, suggesting the Dice objective favors entity detection over boundary accuracy.
- **Focal Loss** severely degrades precision ($P = 0.8063$), indicating it over-emphasizes recall ($R = 0.8869$) at the expense of precision, likely due to aggressive down-weighting of easy examples.
- **CRF + Focal Loss** falls between CRF and Focal Loss in precision ($P = 0.8982$), inheriting some precision loss from Focal Loss despite CRF's structural constraints.

Convergence and Stability. All variants exhibit smooth, monotonic improvement in $F1$ across epochs, with convergence achieved by epoch 3–4. **CE + Dice Loss** demonstrates the most dramatic jump in $F1$ from epoch 1 (0.8880) to epoch 2 (0.9050), indicating efficient learning from the combined loss signals. **Focal Loss** and **Dice Loss** show slower convergence and lower epoch-wise $F1$ values throughout training, suggesting these objectives require more careful tuning or are fundamentally misaligned with the task structure.

Validation loss patterns confirm stability: CE + Dice, Base CE, CRF, and CRF + Focal Loss all stabilize after epoch 2 with minimal fluctuation, whereas Focal Loss maintains the lowest absolute validation loss but yields the worst $F1$, indicating a disconnect between loss minimization and metric maximization.

Conclusion. DistilBERT with CE + Dice Loss represents the optimal configuration among the tested variants, achieving state-of-the-art performance on this NER task with $F1 = 0.9109$, precision = 0.9066, recall = 0.9152, and token accuracy = 97.46%. The narrow performance gap between the top variants (CE + Dice, Base CE, CRF) underscores that the baseline is already highly effective, leaving limited room for improvement via advanced techniques. More sophisticated losses and decoding strategies contribute only marginal gains, and in the case of Focal Loss, degrade performance. These findings suggest that the NER task at hand exhibits characteristics favorable to standard approaches: relatively balanced class distribution, clean label sequences, and sufficient model capacity in DistilBERT to capture linguistic patterns without special handling of imbalance or sequence constraints.

4.5 Model Comparison and Discussion

4.5.1 Cross-Architecture Performance Comparison

Table 23: All Model Variants Across All Architectures (Final Epoch Results).

Architecture	Loss Function	Params	F1-score	Precision	Recall	Accuracy
BERT-base (110M Parameters)						
BERT-base	CE + Dice Loss	110M	**0.9202**	0.9183	0.9222	0.9759
BERT-base	Base CE	110M	0.9172	0.9140	0.9204	0.9755
BERT-base	CRF	110M	0.9137	0.9082	0.9194	0.9747
BERT-base	CRF + Focal Loss	110M	0.9118	0.9068	0.9169	0.9746
BERT-base	Dice Loss	110M	0.8959	0.8983	0.8935	0.9686
BERT-base	Focal Loss	110M	0.8502	0.8121	0.8922	0.9589
RoBERTa-base (125M Parameters)						
RoBERTa-base	CE + Dice Loss	125M	**0.9223**	**0.9215**	0.9231	0.9772
RoBERTa-base	CRF	125M	0.9220	0.9195	**0.9246**	**0.9772**
RoBERTa-base	CRF + Focal Loss	125M	0.9203	0.9172	0.9234	0.9769
RoBERTa-base	Base CE	125M	0.9185	0.9157	0.9212	0.9769
RoBERTa-base	Dice Loss	125M	0.9090	0.9125	0.9055	0.9725
RoBERTa-base	Focal Loss	125M	0.8535	0.8176	0.8926	0.9600
DistilBERT-base (66M Parameters)						
DistilBERT-base	CE + Dice Loss	66M	**0.9109**	**0.9066**	**0.9152**	**0.9746**
DistilBERT-base	Base CE	66M	0.9095	0.9046	0.9145	0.9744
DistilBERT-base	CRF	66M	0.9064	0.9003	0.9126	0.9738
DistilBERT-base	CRF + Focal Loss	66M	0.9049	0.8982	0.9117	0.9736
DistilBERT-base	Dice Loss	66M	0.8911	0.8937	0.8884	0.9685
DistilBERT-base	Focal Loss	66M	0.8447	0.8063	0.8869	0.9572

Key Findings **Best Configuration:** RoBERTa-base + CE + Dice Loss achieves the highest F1-score (0.9223) with balanced precision and recall.

Performance Hierarchy: RoBERTa (125M) → BERT (110M, -0.0021 F1) → DistilBERT (66M, -0.0093 F1). Despite 47% fewer parameters, DistilBERT remains competitive, making it ideal for resource-constrained deployment.

Loss Function Consistency: CE + Dice Loss outperforms alternatives across all architectures, while Focal Loss consistently underperforms (-6–7% F1), indicating the findings are architecture-agnostic.

Precision-Recall Balance: CE + Dice achieves the most balanced profiles, with a slight recall bias (0.004–0.0049 F1 points higher than precision) beneficial for NER applications.

Recommendations

- **Maximum Accuracy:** RoBERTa-base + CE + Dice (F1 = 0.9223)
- **Production (Best Trade-off):** BERT-base + CE + Dice (F1 = 0.9202, 12% fewer parameters)
- **Edge/Latency-Critical:** DistilBERT-base + CE + Dice (F1 = 0.9109, 47% parameter reduction)
- **Loss Function:** Always use CE + Dice Loss; avoid Focal Loss for balanced NER datasets

Conclusion RoBERTa + CE + Dice Loss is the best-performing configuration (F1 = 0.9223). However, minimal performance gaps among top variants suggest that architecture choice should prioritize deployment constraints (latency, memory, cost) over marginal accuracy differences.

5 Conclusion

5.1 Summary

This project presents a comprehensive study of Transformer-based Named Entity Recognition (NER) for IT recruitment job descriptions. We construct a domain-specific dataset of 11,001 synthetic samples with token-level IOB annotations for five practical entity types: **ROLE**, **SKILL**, **LOC**, **EXP**, and **SALARY**. To ensure training quality, we apply a two-stage preprocessing pipeline including generation validation and tokenizer-aware label alignment for subword tokenization.

We evaluate three pre-trained Transformer backbones—**BERT-base**, **RoBERTa-base**, and **DistilBERT-base**—under six optimization settings: Base CE, Focal Loss, Dice Loss, CE + Dice, CRF, and CRF + Focal Loss. Across all architectures, the experimental results consistently show that **CE + Dice Loss** is the most effective objective. The overall best model is **RoBERTa-base + CE + Dice**, achieving an F1-score of **0.9223** with strong precision-recall balance and high token-level accuracy. BERT-base + CE + Dice follows closely (F1 = 0.9202), while DistilBERT-base + CE + Dice remains highly competitive (F1 = 0.9109) with substantially fewer parameters.

These findings demonstrate that hybrid loss design can outperform standalone objectives, and that model selection should consider both predictive quality and deployment constraints (latency, memory, and compute budget).

5.2 Limitations

Despite strong performance, this study has several limitations.

First, the dataset is **synthetic** and generated by an LLM. Although diversity constraints are imposed, generated text may not fully reflect the linguistic noise, ambiguity, and formatting irregularities found in real-world job postings. This can limit external validity.

Second, experiments are conducted on a **single domain** (IT recruitment) with a fixed entity schema. The conclusions may not transfer directly to other recruitment sectors (e.g., healthcare, finance) or broader information extraction tasks.

Third, we report results from a fixed train/test split and limited hyperparameter settings. While this supports fair model comparison, it does not fully capture variance across random seeds, data splits, or broader hyperparameter search spaces.

Fourth, the work primarily emphasizes model-level metrics (F1, precision, recall, accuracy) and does not include deeper **error taxonomy** analyses such as boundary errors, type confusion matrices, or robustness under noisy/abbreviated inputs.

Finally, the current study focuses on offline evaluation and does not include production-oriented measurements such as end-to-end latency under realistic traffic, drift monitoring, or human-in-the-loop correction workflows.

5.3 Future Work

Several promising directions can extend this work.

- **Real-world data integration:** Combine synthetic data with manually annotated real job descriptions to improve robustness and reduce synthetic-to-real domain gap.
- **Cross-domain generalization:** Evaluate transfer performance to non-IT recruitment domains and multilingual settings.
- **Advanced modeling strategies:** Explore modern encoder families (e.g., DeBERTa variants), parameter-efficient fine-tuning (LoRA/adapters), and span-based or prompt-based NER formulations.

- **Improved imbalance handling:** Investigate alternative reweighting and calibration strategies (e.g., class-balanced loss, label smoothing, threshold tuning) given the consistent weakness of Focal Loss in this setting.
- **Data quality and active learning:** Introduce uncertainty-driven sampling and iterative annotation to prioritize informative examples and reduce labeling cost.
- **Deployment and MLOps evaluation:** Benchmark inference speed, memory usage, and throughput across hardware targets; design monitoring pipelines for data drift and model degradation.
- **Explainability and trust:** Add token-level attribution and error explanation tools to support recruiter-facing transparency and easier model debugging.

In conclusion, this work confirms that Transformer-based NER is highly effective for extracting structured recruitment information from unstructured job descriptions. Among the evaluated strategies, **RoBERTa-base with CE + Dice Loss** offers the best raw performance, while **DistilBERT with CE + Dice Loss** provides an attractive efficiency-performance trade-off for practical deployment.

References

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A Complete Model Comparison

Table 24: All Model Variants Across All Architectures (Final Epoch Results).

Architecture	Loss Function	Params	F1-score	Precision	Recall	Accuracy
BERT-base (110M Parameters)						
BERT-base	CE + Dice Loss	110M	**0.9202**	0.9183	0.9222	0.9759
BERT-base	Base CE	110M	0.9172	0.9140	0.9204	0.9755
BERT-base	CRF	110M	0.9137	0.9082	0.9194	0.9747
BERT-base	CRF + Focal Loss	110M	0.9118	0.9068	0.9169	0.9746
BERT-base	Dice Loss	110M	0.8959	0.8983	0.8935	0.9686
BERT-base	Focal Loss	110M	0.8502	0.8121	0.8922	0.9589
RoBERTa-base (125M Parameters)						
RoBERTa-base	CE + Dice Loss	125M	**0.9223**	**0.9215**	0.9231	0.9772
RoBERTa-base	CRF	125M	0.9220	0.9195	**0.9246**	**0.9772**
RoBERTa-base	CRF + Focal Loss	125M	0.9203	0.9172	0.9234	0.9769
RoBERTa-base	Base CE	125M	0.9185	0.9157	0.9212	0.9769
RoBERTa-base	Dice Loss	125M	0.9090	0.9125	0.9055	0.9725
RoBERTa-base	Focal Loss	125M	0.8535	0.8176	0.8926	0.9600
DistilBERT-base (66M Parameters)						
DistilBERT-base	CE + Dice Loss	66M	**0.9109**	**0.9066**	**0.9152**	**0.9746**
DistilBERT-base	Base CE	66M	0.9095	0.9046	0.9145	0.9744
DistilBERT-base	CRF	66M	0.9064	0.9003	0.9126	0.9738
DistilBERT-base	CRF + Focal Loss	66M	0.9049	0.8982	0.9117	0.9736
DistilBERT-base	Dice Loss	66M	0.8911	0.8937	0.8884	0.9685
DistilBERT-base	Focal Loss	66M	0.8447	0.8063	0.8869	0.9572